

Vaishnavi Thirumala

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PROFESSIONAL SUMMARY

High-achieving Artificial Intelligence & Data Science student specializing in building production-grade MLOps pipelines and Generative AI architectures. Proven expertise in engineering automated retraining pipelines, multi-hop retrieval systems, and stateful agent workflows.

EDUCATION

Indian Institute of Information Technology, Sri City
Bachelor of Technology – Artificial Intelligence and Data Science
CGPA: 9.52

Chittoor, AP
Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, R (Familiarity)
ML/AI: PyTorch, TensorFlow, Scikit-learn, NLP, Reinforcement Learning
Gen AI & LLMs: LangChain, LangGraph, RAG, Retrieval Processes, Gemini API, Vector DBs
Data & Tools: Pandas, NumPy, SciPy, MLflow, DVC, Git
Backend/Systems: FastAPI, Docker, Streamlit
Analytics: SHAP/LIME, A/B Testing, Bayesian Inference

PROJECTS

ChurnOps: Automated MLOps Pipeline for Churn Prediction

[GitHub](#)

- Engineered a production-grade stacking ensemble (XGBoost, LightGBM, Random Forest + LR meta-learner via 5-fold CV), boosting accuracy from **81.2% to 90.4%** and ROC-AUC from 0.84 to **0.91** on the IBM Telco dataset (7,043 rows).
- Designed a business-aware decision threshold optimizer evaluating 100 probability intervals against customer LTV (\$500) and campaign cost (\$50), maximizing net profit to **~\$40,800 per 1,000 customers (560%+ ROI)**.
- Built a closed-loop, self-healing retraining system using **Evidently AI** for drift detection, **Prometheus** for metric exposure, and **GitHub Actions** to auto-trigger DVC retraining and MLflow model promotion when drift share exceeds 50%.
- Tech:** *XGBoost, LightGBM, Optuna, SHAP, DVC, MLflow, Evidently AI, Prometheus, Grafana, Docker*

Multi-Hop RAG Fact Verification System

[GitHub](#)

- Developed an end-to-end multi-hop RAG pipeline verifying claims against a Wikipedia corpus, achieving **66.67% Recall@5** on FEVER benchmark claims with **1.2s average latency on CPU-only hardware**.
- Engineered a 2-hop iterative retrieval engine combining **BM25 Okapi** sparse search with dynamic in-memory **FAISS** dense reranking (*all-MiniLM-L6-v2*, 384-dim vectors), keeping the full pipeline under **<1GB RAM** with zero cloud dependency.
- Integrated **spaCy NER** for dynamic query expansion across entity types (PERSON, ORG), enabling multi-hop bridging across disjoint documents and feeding a *cross-encoder/nli-distilroberta-base* model for three-class verdict classification.
- Tech:** *BM25Okapi, FAISS, SentenceTransformers, spaCy, HuggingFace, FastAPI, Pyvis, NetworkX*

Personalized Email Campaign Generator with A/B Testing

[Live](#) | [GitHub](#)

- Orchestrated a stateful LangGraph agent workflow across multi-LLM providers (Gemini, GPT-4o, Claude) to generate persona-conditioned email variants per CRM segment (*young_professional, executive, retiree*, etc.) with structured JSON output parsing.
- Implemented a Bayesian A/B testing engine using **Beta-distribution conjugate priors** and **Monte Carlo simulation** ($N = 100,000$) to evaluate $P(\text{CTR}_A > \text{CTR}_B)$, automatically declaring a winner at a **95% confidence threshold**.
- Built a closed-loop prompt optimization system appending winning variant attributes to a dynamic few-shot JSON bank per segment, iteratively steering the LLM toward higher-performing copy styles across up to 5 optimization loops.
- Tech:** *LangGraph, LangChain, Gemini API, GPT-4o, Claude API, FastAPI, SciPy, MLflow, Streamlit, Docker*

LEADERSHIP & EXPERIENCE

NLP Domain Lead
Epoch – AI/ML Club of IIT Sri City
Design Team Core Member
IIT Sri City

Aug 2025 – April 2026
Sri City, AP
Oct 2024 – Aug 2025
Sri City, AP